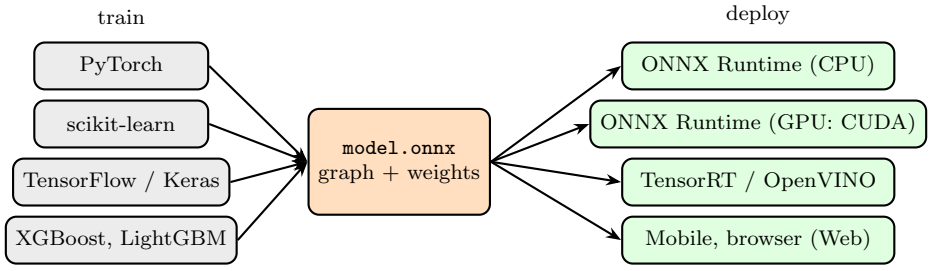
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ONNX (Open Neural Network Exchange)

What it is: an **open, framework-independent format** to represent trained ML models, created by Microsoft and Facebook in 2017 and now hosted by the Linux Foundation. The idea: **train in any framework, deploy with any runtime**. Without a common format, connecting N frameworks to M runtimes or hardware targets needs $N \times M$ converters; with ONNX as the hub, only $N + M$.



What a `.onnx` file contains: a **computational graph** whose nodes are standard **operators** (`MatMul`, `Conv`, `Relu`, `Softmax`, ...), the **weights** as initializers, typed inputs and outputs, and an **opset version** that fixes the exact meaning of each operator. It is serialized with Protocol Buffers, so it is independent of Python and of the training framework.

ONNX Runtime (ORT): Microsoft's high-performance engine for running ONNX models. It applies graph optimizations (constant folding, operator fusion) and sends work to **execution providers**: CPU, CUDA, TensorRT, OpenVINO (Intel), DirectML (Windows), Core ML (Apple), QNN (Qualcomm). The same file runs on a server GPU, a laptop CPU, a phone or a browser.

```
torch.onnx.export(model, example_input, "model.onnx")                    # export from PyTorch
sess = onnxruntime.InferenceSession("model.onnx", providers=["CUDAExecutionProvider"])
y = sess.run(None, {"input": x_numpy})                                    # inference with ORT
```

Why use it: portability (serving does not need PyTorch or TensorFlow installed, no lock-in), **speed** (ORT is often faster than the training framework in inference mode, with int8 quantization), and **classic ML too** (`skl2onnx`, `onnxmltools` convert scikit-learn, XGBoost and LightGBM, e.g. to serve them from C#, Java or C++).

Limitations: custom or very new operators may be missing from the opset, dynamic control flow and shapes can be hard to export, and outputs should be checked against the original model within a tolerance.